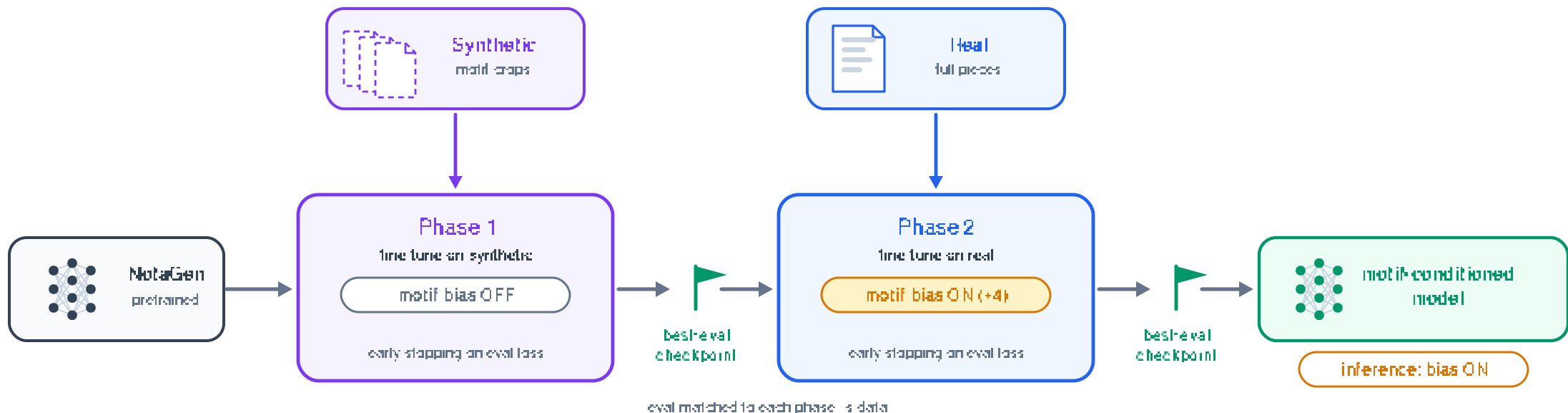
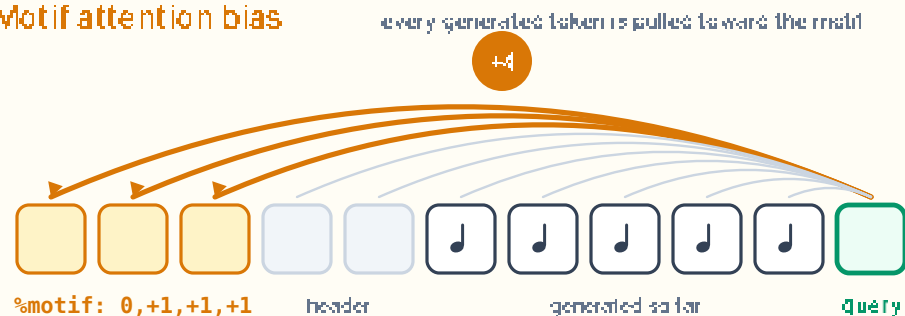
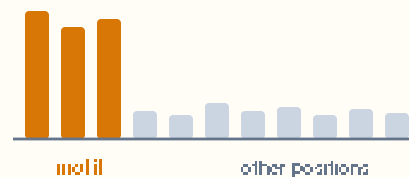




Motif attention bias



attention weights



Attention =

$$\text{softmax}(QK^T/\sqrt{d} + b \cdot 1[\text{motif}])$$

$b = +4$ on motif key positions
all layers before softmax

$b = +4$ optimal $+6/+8$ collapse

Bias applied in Phase 2 training (matched eval and inference) the loss itself stays unweighted cross entropy